

Supplier Discovery Tool

Team Lead: Adam · Team: Data Engineering · Duration: ~6 weeks · Status: Kickoff

The one-line version

Build a small program that **finds potential suppliers online, collects useful information about each one, scores them** so the best ones rise to the top, and outputs a clean list a human can act on.

Why this matters (read this once, then keep it in mind): Finding good suppliers is slow and manual today. If you automate the boring part — searching, collecting, organizing — a human only has to look at a short, ranked list instead of hunting across dozens of websites. You are building the machine that does the hunting.

What you're building, in plain terms

Think of it as a pipeline with four stages. Data goes in one end messy, comes out the other end clean and ranked. You have 6 weeks and two people — build the stages **one at a time** and get stage 1 fully working before touching stage 2.

1 Collect (scrape / aggregate)

Gather a raw list of candidate suppliers from public sources — business directories, marketplace seller pages, category listings. For each one, grab whatever basic info is on the page: name, website, category, location, any contact info shown.

2 Enrich (add useful details)

Take each supplier from stage 1 and fill in the blanks: what category do they sell in, do they have a phone/email/social page, any signals about pricing or size. Enrichment just means "make each row more complete and more useful."

3 Rank (score the best ones)

Give each supplier a score using simple rules (e.g. has contact info = +points, relevant category = +points, more complete data = +points). Sort so the strongest candidates are at the top. Start with basic rules — you can make it smarter later.

4 Output (a clean, ranked list)

Export the final result as a clean spreadsheet or table, sorted best-to-worst, with a **draft outreach message** pre-written for each supplier. **A human reviews and sends the messages — the tool never sends anything automatically.**

What information to collect per supplier

Aim for a table where every supplier is one row with these columns. If a field is missing, leave it blank — that's fine.

Field	What it means
name	Supplier / company name
category	What they sell (electronics, home, beauty...)
website	Their site or store page URL
contact	Phone, email, or WhatsApp if shown publicly
location	City / region
price_signal	Any hint about price range (low / mid / high)
source	Where you found them (which site)
score	The rank score your tool calculates

Suggested tools

Nothing fancy required. Stick to simple, well-documented Python libraries:

- requestsBeautifulSoup
- fetch and read web pages
- pandas
- organize rows and columns, export to spreadsheet

SQLite or a simple CSV to store results (start with CSV, upgrade later)

Pick the simplest thing that works. A messy script that produces a real list beats a clever one that never finishes.

Milestones (roughly one per week)

Week 1 — Setup + planning. Pick 1–2 source sites, agree how you'll split the work between the two of you, and get a basic script fetching one page successfully.

Week 2 — Stage 1. Get scraping working end to end: pull a raw list of ~50 suppliers into a CSV. Just names + whatever is easy to grab.

Week 3 — Stage 2 (part 1). Start enriching: add category, location, and website for each supplier.

Week 4 — Stage 2 (part 2). Fill in contact info and price signals. Aim for mostly-complete rows.

Week 5 — Stage 3. Write the scoring rules, sort the list. Best candidates on top.

Week 6 — Stage 4. Clean export + auto-drafted outreach message per supplier. Polish, write the README.

Definition of done

- I can run one command and get a clean spreadsheet of ranked suppliers.
- Each supplier has most fields filled in and a score.
- Each row has a draft outreach message ready for a human to review and send.
- There's a short README explaining how to run it.
- At least **50 real suppliers** in the final list.

Ground rules (important):

- Only collect **public** information. Respect each site's `robots.txt` and terms of use.
- Go slow — add small delays between requests so you don't hammer a site.
- The tool **drafts** outreach only. It never sends messages automatically. A human always sends.

When you get stuck: this project is yours to own. First check the library docs, search the error, and try a smaller version of the problem. Most answers are in the docs or a quick search. Talk to your teammates before escalating. The goal is for you to solve it — that's the whole point of owning it.